URBAN EDGE COLORFUL YOU

Assignment - 1 Number system

1] Complete the table with equivalents in binary, octal and hexa for: 143.75 (decimal), 11011.101 (binary), 453 (octal), 1F.C (hexa)

Decimal	binary	Octal	Hexa
143.75			
	11011.101		
			1F.C

→ (a) $(143.75)_{10} \rightarrow (?)_2$ decimal → binary

$$\begin{array}{l|l} 2 & 143 \\ 2 & 71 - 1 \\ 2 & 35 - 1 \\ 2 & 17 - 1 \\ 2 & 8 - 1 \\ 2 & 4 - 0 \\ 2 & 2 - 0 \\ & 1 - 0 \end{array}$$

$0.75 \times 2 = 1.5$

$0.5 \times 2 = 1$

$0.75 = .11$

143 = 10001111

$\therefore (143.75)_{10} = (10001111.11)_2$

$(10001111.11)_2 \rightarrow (?)_8$ binary → octal

10001111.11

$\underbrace{010}\ \underbrace{001}\ \underbrace{111}.\underbrace{110}$

217.6

$\therefore (10001111.11)_2 = (217.6)_8$

$(10001111.11)_2 \rightarrow (?)_{16}$ binary → hexa

10001111.11

$\underbrace{1000}\ \underbrace{1111}.\underbrace{1100}$

8F.C

$\therefore (10001111.11)_2 = (8F.C)_{16}$

(b) $(11011.101)_2 \rightarrow (?)_{10}$ binary → decimal

$= 1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1 . 1 \times 1/2 + 0 \times 1/4 + 1 \times \frac{1}{8}$

$= 16 + 8 + 0 + 2 + 1 . 0.5 + 0 + 0.125$

$= 27.625$

$\therefore (11011.101)_2 = (27.625)_{10}$

$(11011.101)_2 \rightarrow (?)_8$ binary → octal

11011.101

$\underbrace{011}\ \underbrace{011}.\underbrace{101}$

33.5

$\therefore (11011.101)_2 = (33.5)_8$

Scanned with OKEN Scanner

$(11011.101)_2 \rightarrow (?)_{16}$ binary → hex

11011.101

$\underbrace{0001}\underbrace{1011}.\underbrace{1010}$

1B.A

$\therefore (11011.101)_2 = (1B.A)_{16}$

(c) $(453)_8 \rightarrow (?)_{10}$ octal → decimal

$= 4\times8^2 + 5\times8^1 + 3\times8^0$

$= 256 + 40 + 3$

$= 299$ $\therefore (453)_8 = (299)_{10}$

$(453)_8 \rightarrow (?)_2$ octal → binary

$4 \rightarrow 100$

$5 \rightarrow 101$

$3 \rightarrow 011$

100 101 011 $\therefore (453)_8 = (100101011)_2$

$(100101011)_2 \rightarrow (?)_{16}$ binary → hex

100101011

$\underbrace{0001}\ \underbrace{0010}\ \underbrace{1011}$

12B $\therefore (100101011)_2 = (12B)_{16}$

(d) $(1F.C)_{16} \rightarrow (?)_{10}$ hex → decimal

$= 1\times16^1 + 15\times16^0 + 12\times16^{-1}$

$= 31.75$ $\therefore (1F.C)_{16} = (31.75)_{10}$

$(1F.C)_{16} \rightarrow (?)_2$ hex → binary

$1 \rightarrow 0001$

$F \rightarrow 1111$

$C \rightarrow 1100$

00011111.1100

11111.11 $\therefore (1F.C)_{16} = (11111.11)_2$

$(11111.11)_2 \rightarrow (?)_8$

11111.11 binary → octal

$\underline{011}\ \underline{111}.\ \underline{110}$

37.6 $\therefore (11111.11)_2 = (37.6)_8$

Final table

Decimal	Binary	octal	hex
143.75	10001111.11	217.6	8F.C
27.625	11011.101	33.5	1B.A
299	100101011	453	12B
31.75	11111.11	37.6	1F.C

2) Find equivalents

(a) $(324)_5 = (?)_{10}$

$\rightarrow$ $= (3\times5^2)+(2\times5^1)+(4\times5^0)$

$= 75+10+4$

$= 89$

$\therefore (324)_5 = (89)_{10}$

(b) $(32.23)_4 = (?)_{10}$

$\rightarrow$ $= (3\times4^1)+(2\times4^0)+(2\times4^{-1})+(3\times4^{-2})$

$= 12+2+0.5+0.1875$

$= 14.6875$

$\therefore (32.23)_4 = (14.6875)_{10}$

(c) $(231)_{10} = (?)_7$

$\rightarrow$
7 | 231
7 | 33 - 0
4 - 5

450

$\therefore (231)_{10} = (450)_7$

(d) $(189)_{10} = (?)_6$

$\rightarrow$
6 | 189
6 | 31 - 3
5 - 1

$\therefore (189)_{10} = (513)_6$

3) Find the radix r

(a) $\left(\frac{312}{20}\right)_r = (13.1)_r$

$\rightarrow$ $(312)_r = 3r^2+1r+2 = 3r^2+r+2$

$(20)_r = 2r+0 = 2r$

$(13.1)_r = 1r+3+1r^{-1} = r+3+\frac{1}{r}$

$\therefore \frac{3r^2+r+2}{2r} = r+3+\frac{1}{r}$

$3r^2+r+2 = 2r^2+6r+2$

$3r^2-2r^2+r-6r+2-2=0$

$r^2-5r=0$

$r(r-5)=0$

Since, radix can't be zero, $\boxed{r=5}$

(b) $264_r + 136_r = 433_r$

→ $(264)_r = 2r^2+6r+4$

$(136)_r = 1r^2+3r+6$

$(433)_r = 4r^2+3r+3$

$(2r^2+6r+4)+(r^2+3r+6) = (4r^2+3r+3)$

$3r^2+9r+10 = 4r^2+3r+3$

$-3r^2+4r^2-9r+3r-10+3=0$

$r^2-6r-7=0$

$(r-7)(r+1)=0$

Since, radix can't be negative, $\boxed{r=7}$

(c) $(221505)_8 = (12345)_r$

→ $(221505)_8 = (2\times8^5)+(2\times8^4)+(1\times8^3)+(5\times8^2)+(0\times8^1)+(5\times8^0)$

$= 65536+8192+512+320+0+5$

$= 74565$

$(12345)_r = 1r^4+2r^3+3r^2+4r^1+5r^0$

$= r^4+2r^3+3r^2+4r+5$

for $r=16$

$= 16^4+(2\times16^3)+(3\times16^2)+(4\times16)+5$

$= 65536+8192+768+64+5$

$= 74565$

hence, $\boxed{r=16}$

4) Addition

(a) $(1001)_2 + (11011111)_2$

→

URBAN EDGE COLORFUL YOU ___/___/___

$$\begin{array}{r} 00001101 \\ 11011111 \\ \hline 11101100 \end{array}$$

$\therefore (1101)_2 + (11011111)_2 = (11101100)_2$

(b) $(77652)_8 + (76543)_8$

→ 77652

76543+

- $2+3=5$
- $5+4=(9)_{10}=(11)_8 \rightarrow$ write $=1$, carry$=1$
- $6+5+1=(12)_{10}=(14)_8 \rightarrow$ write $=4$, Carry$=1$
- $7+6+1=(14)_{10}=(16)_8 \rightarrow$ write $=6$, Carry$=1$
- $7+7+1=(15)_{10}=(17)_8 \rightarrow$ write $=7$, Carry$=1$
- bring down the final carry 1

$$\therefore \begin{array}{r} 77652 \\ 76543 \\ \hline 176415 \end{array}$$

$\therefore (77652)_8 + (76543)_8 = (176415)_8$

(c) $(578A)_{16} + (ABDE)_{16}$

→ 578A

ABDE +

- $A+E=10+14=(24)_{10}=(18)_{16} \rightarrow$ write$=8$, Carry$=1$
- $8+D+1=8+13+1=(22)_{10}=(16)_{16} \rightarrow$ write$=6$, carry$=1$
- $7+B+1=7+11+1=(19)_{10}=(13)_{16} \rightarrow$ write 3, Carry$=1$
- $5+A+1=5+10+1=(16)_{10}=(10)_{16} \rightarrow$ write$=0$, Carry$=1$
- Bring down the final carry 1.

$$\begin{array}{r} 578A \\ ABDE \\ \hline 10368 \end{array}$$

$\therefore (578A)_{16} + (ABDE)_{16} = (10368)_{16}$

(d) $(234)_6 + (315)_6$

→ 234
315

- $4+5 = (9)_{10} = (13)_6$ → write = 3, carry = 1.
- $3+1+1 = (5)_{10} = (5)_6$ → write = 5, carry = 0
- $2+3 = (5)_{10} = (5)_6$ write = 5, carry = 0

  234
  315
  553

$\therefore (234)_6 + (315)_6 = (553)_6$

5) Binary → Gray code

(a) $(10101)_2$

→ First bit = 1

- $1 \oplus 0 = 1$
- $0 \oplus 1 = 1$
- $1 \oplus 0 = 1$
- $0 \oplus 1 = 1$

- 1st gray bit = 1st binary bit
- Every next gray bit = xor of the current binary bit & previous binary bit

Binary 1 0 1 0 1
Gray 1 1 1 1 1

(b) $(01110)_2$

→ First bit = 0

- $0 \oplus 1 = 1$
- $1 \oplus 1 = 0$
- $1 \oplus 1 = 0$
- $1 \oplus 0 = 1$

Binary: 0 1 1 1 0
Gray = 0 1 0 0 1

6) Gray code → Binary

(a) 01101 (gray)

→ First bit = 0

$0 \oplus 1 = 1$
$1 \oplus 1 = 0$
$0 \oplus 0 = 0$
$0 \oplus 0 = 1$

- 1st binary bit = 1st gray bit
- next binary bit = Previous binary bit xor current gray bit

Gray 0 1 1 0 1
binary 0 1 0 0 1

(b) 10101 (gray)

→ First bit = 1

$1 \oplus 0 = 1$

$1 \oplus 1 = 0$

$0 \oplus 0 = 0$

$0 \oplus 1 = 1$

Gray 1 0 1 0 1

binary 1 1 0 0 1